Large Data Set: OCR MEI (London)

Questions 1–11

Question 1. These exercises were written using LDS8 but should continue to be of use for future London Large Data Set samples. State the following facts about the Large Data Set:

a) How many boroughs of London are listed.
b) How many areas of England are recorded (ignoring categories, Wales, England, England and Wales, and UK).
c) Who defines which boroughs are Inner or Outer London for this data set?

Question 2. Explain what each of the following categories in the Large Data Set are:

a) Employment Rate (age 16 to 64).
b) Mean and median Income of Taxpayers (£).
c) All pupils at end of KS4 achieving 5+ A*-C including English and Maths.
d) Household Waste Recycling Rate.

Question 3. Which properties are excluded from the median house price category?

Question 4. Why should the data related to the mean and median income of taxpayers be treated with caution?

Question 5. A sample of 5 London boroughs are being sampled from the Large Data Set. For each of the following, write down the name of the sampling method used.

a) Each borough is given a number between 1 and 33 then a random number generator is used to select 5, until 5 unique boroughs have been identified.

b) The order of the boroughs is randomised, then every 6th is noted.

c) The order of the boroughs is randomised, then the first 5 are sampled.

d) The data is sorted into inner and outer London and then 1 inner London and 4 outer London boroughs are randomly selected using a similar approach to in part a).

Question 6. The table below a sample of 10 boroughs in London and their median house price in 2021:

Area	Inner or Outer London	Median House Price (£)
Barnet	Outer	575 500
Brent	Outer	525 000
Camden	Inner	765 000
Ealing	Outer	522 000
Hammersmith and Fulham	Inner	780 000
Harrow	Outer	535 000
Kensington and Chelsea	Inner	1 250 000
Merton	Outer	545 000
Richmond upon Thames	Outer	715 000
Wandsworth	Inner	656 178

a) Find the mean and the standard deviation of the median house price.

b) Give a reason why the sample used above may have biased data.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

c) Identify any boroughs which are outliers.

The same boroughs in 2007 for median house price x gives

$$\bar{x} = £566\,727.50 \text{ and } s_x = £176\,435.80$$

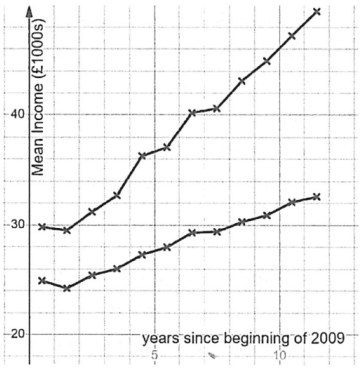
d) Compare these to the statistics calculated above.

Question 7. The mean income of taxpayers and median income of taxpayers in 2018/19 for Lewisham and for Kensington and Chelsea are shown in the table below:

	Mean (£)	Median (£)
Kensington and Chelsea	173 000	39 300
Lewisham	38 000	27 900

Compare the distributions of the income of taxpayers for both areas.

Question 8. The mean income I in £1000s has been recorded against time t for 12 consecutive years for both Hackney and the North West of England.



a) Identify which of the graphs is Hackney.

The two lines may be modelled by:

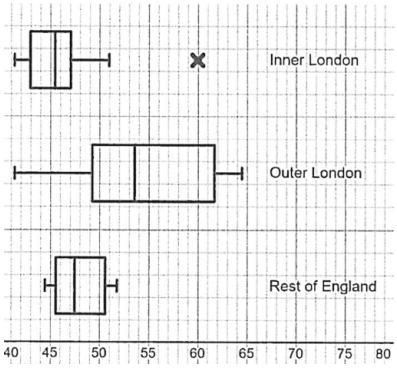
$$I = 1.88t + 26.3 \text{ and } I = 0.781t + 23.3$$

b) Predict the mean income for 2021/2022 according to these models.

c) Explain what the values 1.88 and 26.3, and 0.781 and 23.3, within the models above mean in context.

d) Compare the change in mean income of the two areas.

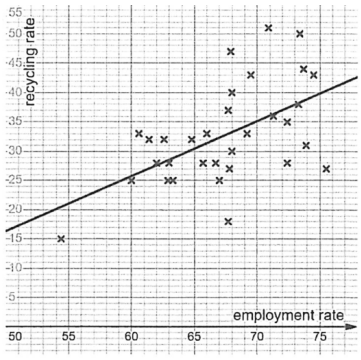
Question 9. The percentage of all pupils at the end of KS4 achieving 5 or more A*- C grades including English and Maths in the academic year 2007/8 for Inner London, Outer London and the rest of England are shown in the boxplots below.



a) Using your knowledge of the Large Data Set, identify the name of the outlier in Inner London.
b) Compare the 3 data sets.

Question 10. The scatter graph below shows the household recycling rate (%) against the employment rate for 16-64 year olds for 20 of the London boroughs in 2011. The product moment correlation coefficient is $r = 0.7227$ and the least squares regression line is given by

$$y = 0.979x - 33.0$$



a) What does the value of r mean for the association between the household recycling rate and the employment rate?

One of the boroughs not included in the graph has an employment rate of 73.4%.

b) Find the recycling rate of this borough according to the regression line.

Lewisham was not included in the sample. It had an employment rate of 67.7% and a recycling rate of 18% in 2011.

c) If this data point were to be included, explain how the regression line and the value of r would change.

Question 11. Prerequisite knowledge for Q11: Normal Distribution The percentage of all pupils achieving 5 or more A*- C grades across the whole of England in 2011/2 may be considered normally distributed, where $X \sim \mathrm{N}(59.4, 4)$. Find

a) the probability that the percentage is higher than 58%.
b) the probability that the percentage is less than 55%.
c) $\mathrm{P}(\mu - 2\sigma < X < \mu + 2\sigma)$